

PCB DESIGN TASK REPORT:

TriComm Adapter

*Design and Development of a USB-Based Multi-Protocol
Communication Interface for UART, SPI and I²C using eSim v2.5*

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Project Overview:

Design and validation of a four-layer USB communication interface that converts a single USB connection into independent UART, SPI, and I²C interfaces using a USB 2.0 hub architecture, regulated 3.3V power management, and high-speed PCB design practices.

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Chapter 1: Introduction

The TriComm Adapter is a compact four-layer embedded communication interface designed to bridge a single USB host connection to three widely used serial communication protocols: UART, SPI, and I²C. The project was developed as a PCB design exercise with emphasis on high-speed USB routing, multi-protocol communication, power integrity, and manufacturable hardware design. Taking inspiration from Northwest Engineering Solutions' USB to UART over RJ-45 interface, this design extends the concept into a versatile multi-interface development module capable of supporting embedded debugging, peripheral evaluation, and hardware prototyping from a single USB connection.

The USB2513BI-AEZG serves as the central hub of the system whose three-port USB 2.0 high speed controller distributes the upstream USB connection to dedicated protocol conversion devices. It receives USB differential data lines from the USB Type A connector and transfers each downstream interface independently, allowing simultaneous access to multiple communication channels through a single physical connection.

The CP2102-GM USB to UART bridge provides a reliable asynchronous serial interface for console communication, firmware programming, and debugging applications. Its transmit and receive signals are routed directly to an RJ-45 connector, enabling long distance UART communication over structured Ethernet cabling while maintaining the electrical characteristics of standard 3.3V TTL serial logic.

To expand the functionality beyond UART, the design integrates the CP2130-F01-GMR USB to SPI bridge and the CP2112-F03-GMR USB to I²C bridge. The CP2130 converts USB packets into a full duplex SPI master interface, exposing MOSI, MISO, SCK, CS, 3.3V, GND, and auxiliary control signals through an 8-pin header for direct connection to SPI sensors, memories, displays, and microcontrollers. Similarly, the CP2112 provides a complete USB controlled I²C master interface through a 4-pin header carrying SDA, SCL, 3.3V, and GND, making the module suitable for rapid evaluation of I²C peripherals without requiring an external microcontroller.

Power management is implemented using the AP2112K-3.3TRG1 low dropout voltage regulator, which converts the incoming USB VBUS (5 V) into a stable 3.3V supply for all digital integrated circuits and external communication headers. The regulator offers low output noise, load regulation, and sufficient current capability for the hub and bridge devices. Distributed decoupling capacitors were placed adjacent to every power pin, while bulk capacitors were incorporated to stabilize transient current demands and improve overall supply integrity.

The adapter was designed as a four-layer board consisting of dedicated signal routing layers together with continuous power and ground planes to reduce return path impedance and electromagnetic interference. Design rules were established for USB differential pair matching, minimum clearance, controlled via usage, polygon-based ground distribution, and copper balancing. Electrical Rule Check (ERC) and Design Rule Check (DRC) were successfully completed with zero violations, confirming schematic consistency and fabrication readiness.

Chapter 2: Schematic Design

The TriComm Adapter schematic is organized into five sheets, each representing a subsystem of the communication interface. The design follows the reference implementation of the Northwest Engineering USB to UART via RJ45 module while extending its functionality through the integration of a Microchip USB 2.0 hub and Silicon Labs USB to SPI, I²C bridge controllers. The complete system converts a single USB host connection into independent UART, SPI, and I²C interfaces operating at 3.3V logic. Design aspects such as USB signal integrity, regulated power distribution, ESD protection, decoupling, and compliance with the recommendations provided in the component datasheets were given special consideration.

2.1 Sheet 1: USB to UART via RJ-45 Interface

This sheet implements the primary USB to UART communication channel using the CP2102N-A02-GQFN28 USB bridge. The upstream USB interface carries the differential D+ and D- signals together with the 5 V VBUS supply, while the CP2102 converts USB packets into standard 3.3V UART signals. The UART transmit and receive lines are routed to an RJ-45 connector, allowing reliable serial communication over twisted pair Ethernet cable for embedded programming, debugging, and industrial control applications.

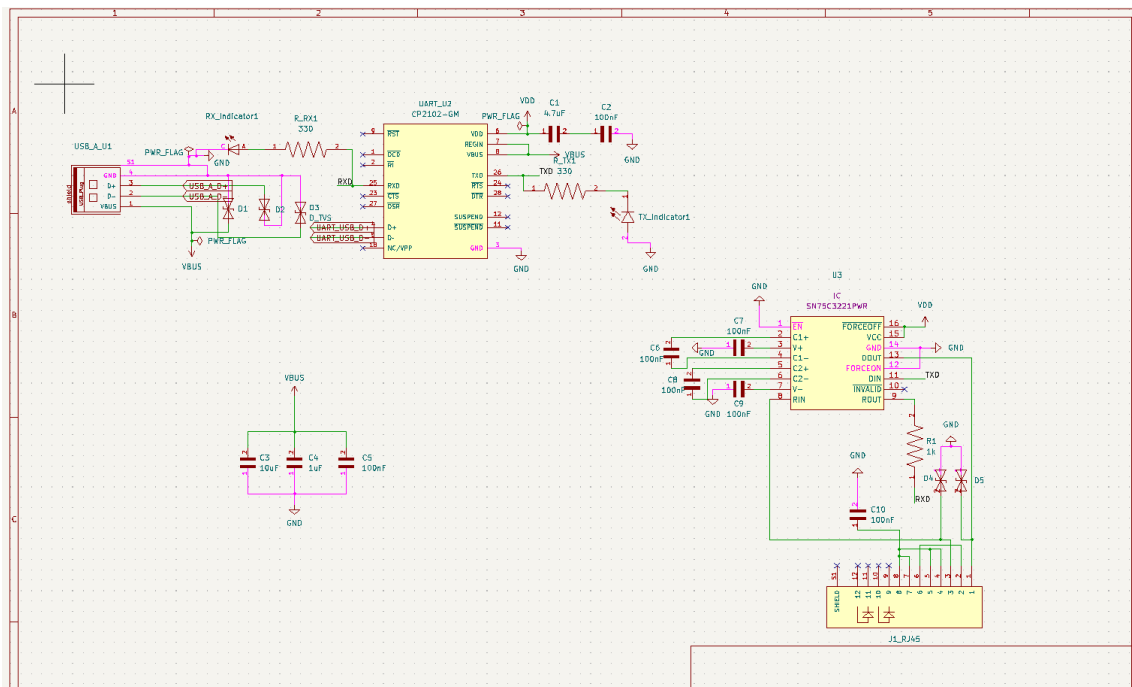


Figure 2.1 USB to UART via RJ45 Interface Schematic

The USB differential pair is routed as a matched impedance signal pair to minimize skew and maintain USB Full-Speed signal quality. Since the CP2102 contains the required USB transceiver and internal pull-up circuitry, no external termination resistors are required on the data lines. The device is powered from the regulated 3.3V rail, ensuring logic compatibility with modern microcontrollers and development boards.

Protection and reliability features

- Low capacitance TVS diodes are connected directly across the USB D+ and D- lines to protect the interface against electrostatic discharge (ESD) events without significantly increasing line capacitance.
- A TVS diode on the VBUS line protects the board from voltage transients introduced through the USB connector.
- Local 100 nF decoupling capacitors are placed adjacent to every CP2102 supply pin to suppress high frequency switching noise and stabilize the internal USB's physical layer during data transmission.

2.2 Sheet 2: USB2513BI-AEZG Hub

The USB hub forms the central communication architecture of the TriComm Adapter. The USB2513BI-AEZG receives one upstream USB connection and distributes it into three independent downstream USB channels, enabling simultaneous operation of the UART, SPI, and I²C bridge devices. The hub relies on the hardware to handle USB setup, data routing, address tracking, and port controls.

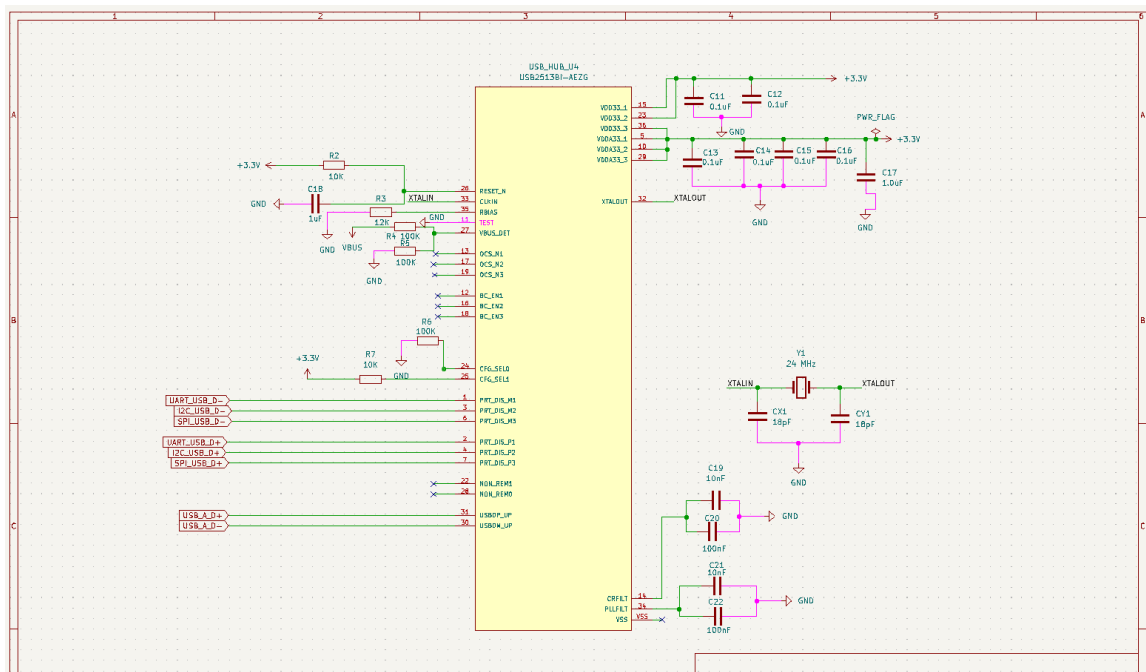


Figure 2.2 USB Hub Schematic

It operates from a 24 MHz crystal oscillator, which provides the timing reference required by the internal USB 2.0 PHY. The configuration straps define the operating mode during startup, while the RBIAS precision resistor establishes the internal analog bias currents. As per Microchip's datasheet, all analog and digital 3.3V supply pins are individually decoupled to maintain stable operation during high-speed USB activity.

Protection and reliability features

- Each VDDA33 pin includes a dedicated 100 nF capacitor, while a shared 1 μ F bulk capacitor supports transient current demand across the analog supply network.
- The exposed thermal pad and all ground pins are connected directly to a continuous ground plane to provide a low impedance return path and improve EMI performance.
- USB differential pairs are routed directly without branching, allowing TVS protection devices to be placed beside the connector as recommended for high-speed USB layouts.

2.3 Sheet 3: USB to SPI Interface (CP2130-F01-GMR)

This section provides a USB controlled SPI Master interface using the CP2130-F01-GMR. The device receives USB commands from one downstream port of the USB2513 hub and generates standard SPI signals including MOSI, MISO, SCLK, and multiple Chip Select outputs. These signals are transferred through an 8-pin header, allowing direct connection to SPI peripherals such as Flash memory, sensors, ADCs, DACs, and display controllers.

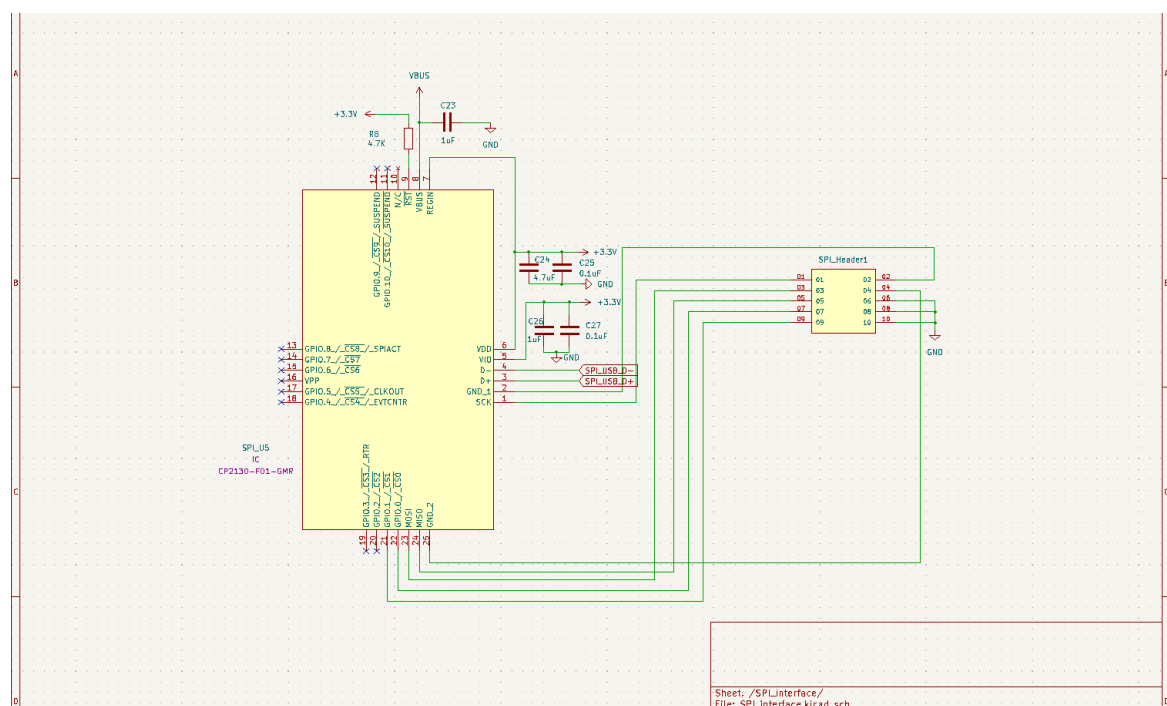


Figure 2.3 SPI Interface Schematic

Unlike conventional USB to serial converters, the CP2130 performs protocol conversion internally and supports configurable SPI clock frequencies and multiple slave select outputs without requiring an external microcontroller. The interface therefore functions as a dedicated USB peripheral controller for hardware evaluation and embedded development.

Protection and reliability features

- Individual 100 nF bypass capacitors are placed beside every supply pin to minimize the ripple generated during SPI clock transitions.
- The SPI header includes regulated 3.3V and GND outputs, providing a complete interface for externally connected peripherals while maintaining a common reference ground.
- Short signal routing between the CP2130 and the output header minimizes clock distortion and improves signal integrity at higher SPI frequencies.

2.4 Sheet 4: USB to I²C Interface (CP2112-F03-GMR)

The I²C subsystem is implemented using the CP2112-F03-GMR, which functions as a USB to I²C controller. The device converts USB transactions into two wire serial communication, exposing SDA and SCL through a 4-pin header together with 3.3V power and ground. This enables direct communication with EEPROMs, RTC modules, GPIO expanders, and other I²C peripherals.

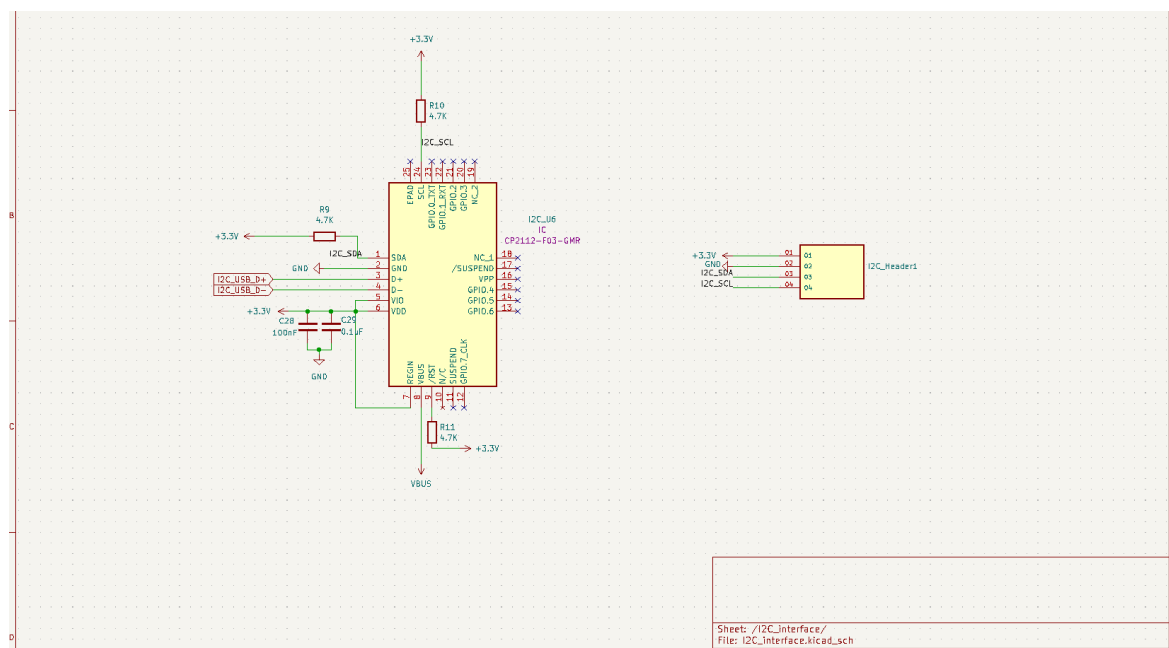


Figure 2.4 I²C Interface Schematic

The CP2112 contains the complete USB interface and I²C controller internally, eliminating firmware development for host side communication. The external pull-up resistors enable the open drain I²C lines to remain high during idle periods, while the USB hub allows the interface to operate independently alongside the UART and SPI channels.

Protection and reliability features

- Precision I²C pull-up resistors ensure proper rise times and reliable bus arbitration on the SDA and SCL lines.
- Local decoupling capacitors suppress digital switching noise produced by USB communication and maintain stable controller operation.
- The dedicated ground reference provided through the header improves communication reliability with externally connected I²C devices.

2.5 Sheet 5: Power Management (AP2112K-3.3TRG1)

The power management circuit converts the incoming 5 V USB VBUS supply into a regulated 3.3V rail using the AP2112K-3.3TRG1 low dropout linear regulator. This regulates supply powers the USB hub, the CP2102 UART bridge, the CP2130 SPI controller, the CP2112 I²C controller, and all external communication headers. The AP2112 was selected due to its low dropout voltage, excellent line and load regulation, low output noise, and current capability suitable for USB powered embedded systems.

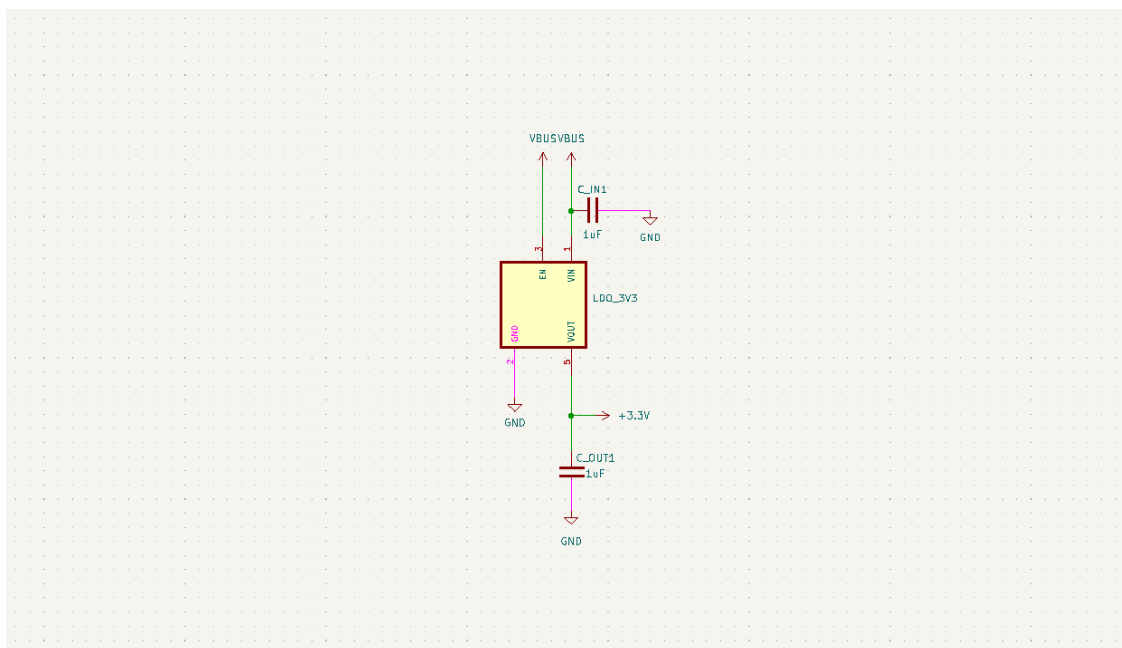


Figure 2.5 Power Management Schematic

The regulator requires minimal external components, making it suitable for compact PCB layouts while still providing stable regulation across varying USB load conditions. Input and output capacitors are selected according to the manufacturer's stability recommendations to ensure reliable transient response and prevent oscillation.

Protection and reliability features

- An input bulk capacitor filters voltage fluctuations present on the USB VBUS line and supplies instantaneous current during load transients.
- The output capacitor maintains regulator stability, reduces output ripple, and improves dynamic load response during simultaneous operation of multiple USB bridge devices.
- Distributed 100 nF decoupling capacitors throughout the PCB isolate individual ICs from supply noise and reduce high frequency current loops, improving overall power integrity.

2.6 Electrical Rules Check

The completed schematic successfully passed Electrical Rules Check (ERC) validation with zero errors or warnings. This verification confirms correct pin classifications, ensures all inputs are properly driven, and validates that net connections align with the component library definitions.

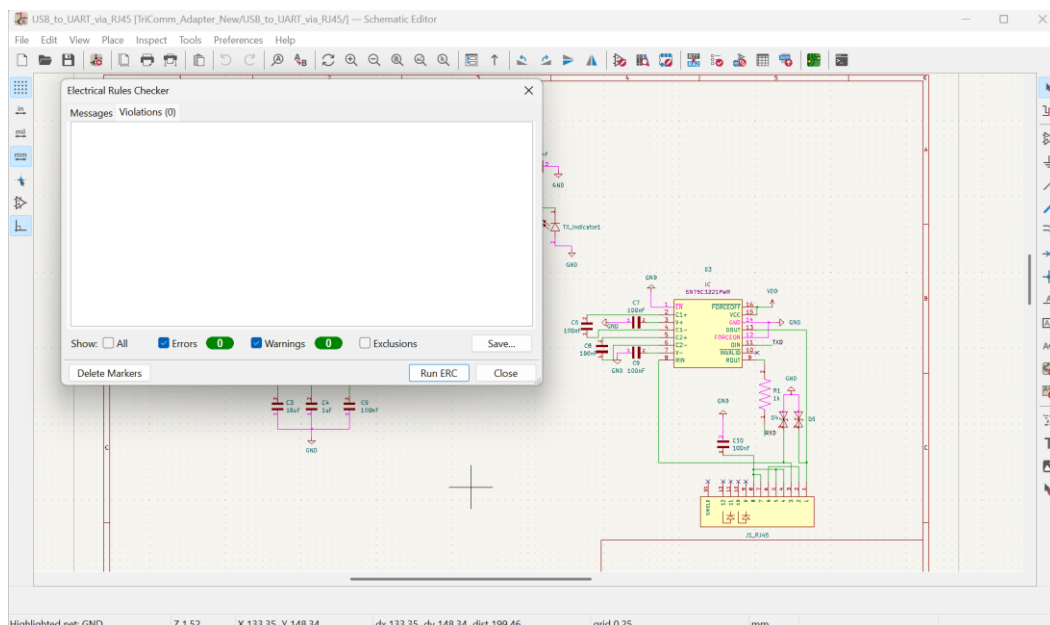


Figure 2.6 Electrical Rules Check Validation

The design of the TriComm Adapter successfully integrates USB to UART, SPI, and I²C communication into a single, USB powered hardware platform using a modular five sheet architecture. By combining the USB2513B hub with various bridge controllers and a regulated 3.3V power supply, the design provides simultaneous multi-protocol communication while maintaining electrical compatibility with modern embedded systems. The inclusion of ESD protection, proper decoupling, bulk capacitance, and recommended USB design practices ensures stable operation, improved signal integrity, and reliable power distribution.

2.7 Footprint Assignment

After completing the schematic, PCB footprints were assigned to every component using the Footprint Editor and Assign Footprints feature. The major integrated circuits were mapped to surface-mount packages matching the exact land patterns and mechanical dimensions from their datasheets. The generic passive components and protection devices were assigned standard SMD footprints to maintain consistency and manufacturability across the board.

The primary communication devices, i.e. the USB2513BI-AEZG, CP2102-GM, CP2130-F01-GMR, CP2112-F03-GMR, and AP2112K-3.3TRG1, were imported using their corresponding QFN, QFN-24, and SOT-23-5 surface mount packages as specified in the manufacturer datasheets. Special attention was paid to the exposed thermal pads on the QFN devices, ensuring a low-impedance ground connection to handle heat dissipation and maintain signal integrity.

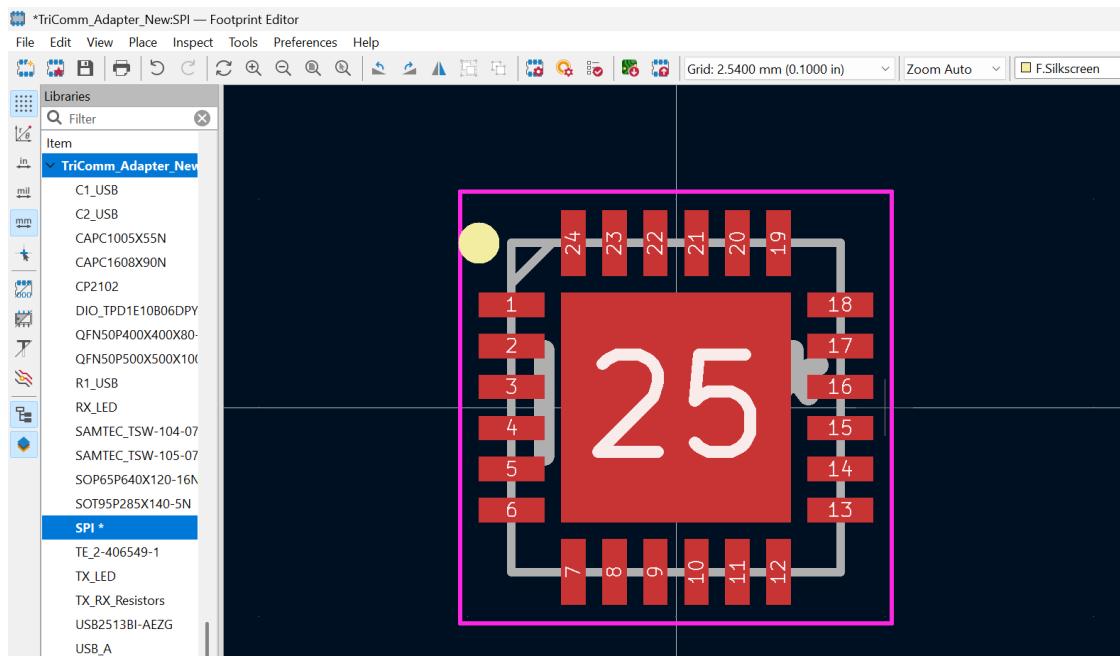


Figure 2.7 Imported Footprint Library

A few supporting components such as ceramic decoupling capacitors, bulk capacitors, resistors, pull-up resistors, LEDs, and TVS diodes were assigned standard SMD footprints from the official footprint libraries. The capacitors and resistors utilized the 0402 and 0603 packages depending on the electrical requirement, while the USB protection diodes employed low capacitance SMD packages suitable for high-speed USB differential signalling. The connectors including the USB Type-A port, RJ-45 connector, 8-pin SPI header, and 4-pin I²C header were assigned manufacturer-compatible footprints to ensure accurate mechanical alignment during PCB assembly.

This complete footprint assignment established a strict one-to-one link between the schematic symbols and physical layout packages. This prepared the design for effective netlist generation, accurate 3D modelling, and smooth DRC validation during the layout phase.

Chapter 3: PCB Layout & Routing

The PCB layout of the TriComm Adapter was designed as a four-layer board to achieve compact routing, reliable signal integrity, and efficient power distribution. The placement of the USB hub, UART, I²C, and SPI interface ICs was optimized to minimize interconnection lengths while maintaining clear separation between high-speed USB signals and power circuitry. A dedicated ground plane and 3.3V power plane were incorporated to improve return current paths and reduce voltage drop across the board.

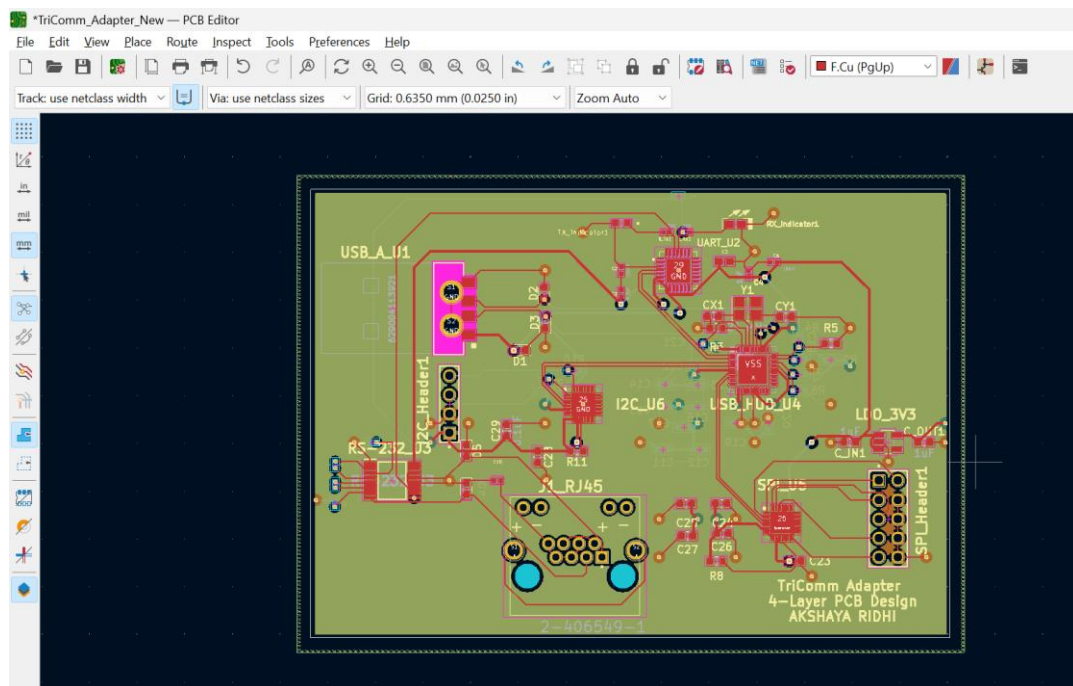


Figure 3.1 PCB Layout

3.1 Routing Methodology

The routing process began with the placement of all major components according to functional blocks, followed by routing of critical signal nets before power distribution. USB differential pairs were routed with parallel traces and consistent spacing to preserve signal integrity between the USB Type-A connector and the USB2513B hub. SPI, I²C, UART, and GPIO signals were then completed using short and direct trace paths to reduce unnecessary routing complexity.

The power routing was implemented using a combination of copper planes and local interconnections. The four-layer stack-up adopted for the design is given below.

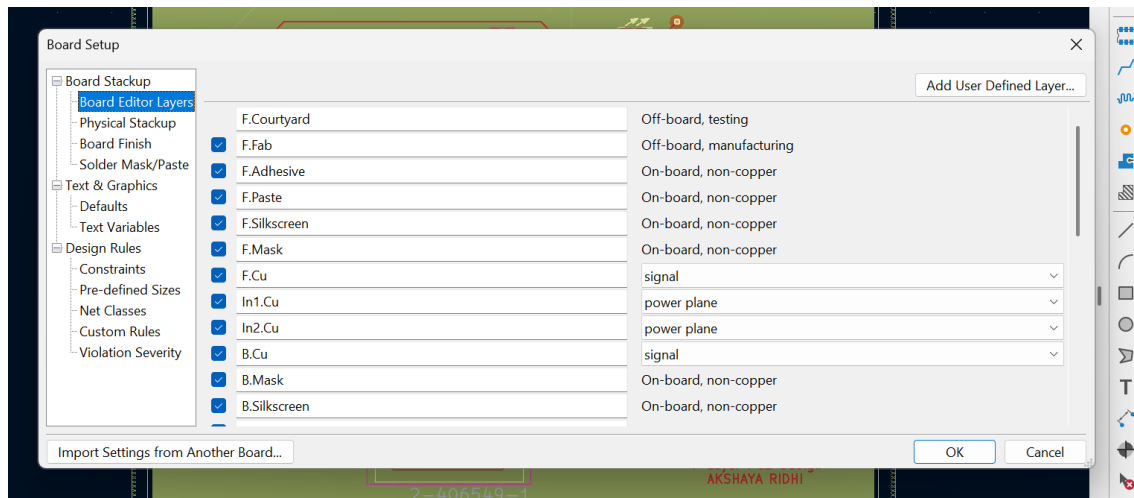


Figure 3.2 Board Layers

Ground connections from surface-mount devices were transferred to the internal GND plane through stitching vias placed adjacent to the component pads. Similarly, all 3.3V supply pins were connected to the dedicated internal power plane using short traces and vias, resulting in a clean and low-impedance power distribution network.

3.2 Design Considerations

Several PCB design practices were incorporated to improve electrical performance and manufacturability.

- A dedicated In2.Cu ground plane provides a continuous return path for all digital signals and minimizes electromagnetic interference.
- The In1.Cu 3.3V plane distributes regulated power uniformly to the USB hub and interface ICs, reducing voltage drop and simplifying routing.
- High-speed USB differential pairs were routed with matched geometry and minimal bends to preserve impedance consistency.
- Decoupling capacitors were positioned immediately adjacent to the power pins of each IC to reduce supply noise and improve transient response.
- Signal traces were kept as short as practical, with power and data routing separated wherever possible to minimize coupling.
- Ground stitching vias were used around power devices and interface ICs to improve plane connectivity and current return paths.

These considerations resulted in a compact layout while maintaining adequate spacing between signal, power, and mechanical features.

Chapter 4: Board Setup

The Board Setup stage established the physical, electrical, and manufacturing constraints required for implementing the schematic as a four-layer PCB. The board configuration was defined to provide adequate signal integrity for the USB differential interfaces, low impedance power and ground distribution, and consistent routing rules throughout the design. The physical stackup, design constraints, and net classes were configured before routing so that the PCB layout remained consistent with the electrical requirements of the circuit.

4.1 Physical Stackup

A four-layer PCB stack up was selected, consisting of two external copper layers and two internal copper layers. The copper layers are separated by FR-4 dielectric material using a combination of prepreg and core construction, providing the mechanical strength and electrical isolation required for multilayer PCB fabrication.

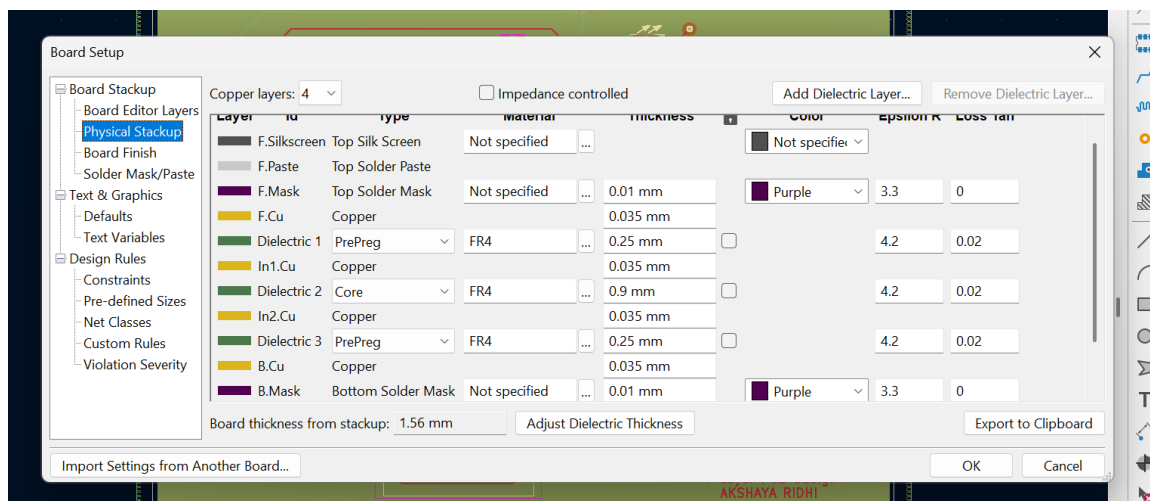


Figure 4.1 Board Physical Stackup

The outer copper layers are primarily used for component placement and signal routing, while the internal layers provide additional routing capacity and continuous copper regions for power and ground distribution. This arrangement also provides controlled and relatively short return paths for high-speed USB signals and helps reduce electromagnetic interference.

4.2 Design Rules Constraints

Design Rule Constraints were established to maintain adequate electrical clearance and ensure that the completed PCB could be reliably fabricated.

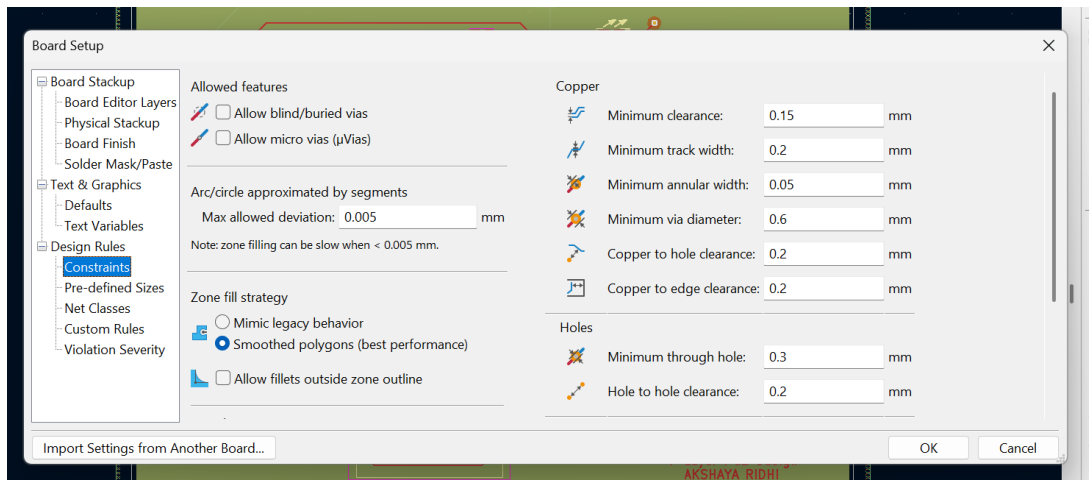


Figure 4.2 Constraints

The rules define minimum clearances between copper features, tracks, pads, vias, and board edges, together with minimum track and via dimensions appropriate for the selected fabrication process. Additional attention was given to the USB differential pairs, where consistent routing geometry, matched trace lengths, and appropriate spacing were maintained to preserve signal integrity. These constraints were applied throughout the layout and verified using Design Rule Check.

4.3 Net Classes

Net classes were defined to apply appropriate routing characteristics according to the electrical function of each connection. High-speed USB differential signals were assigned dedicated routing parameters to maintain consistent trace geometry and spacing, while standard digital communication signals such as UART, SPI, and I²C were routed using general-purpose signal rules.

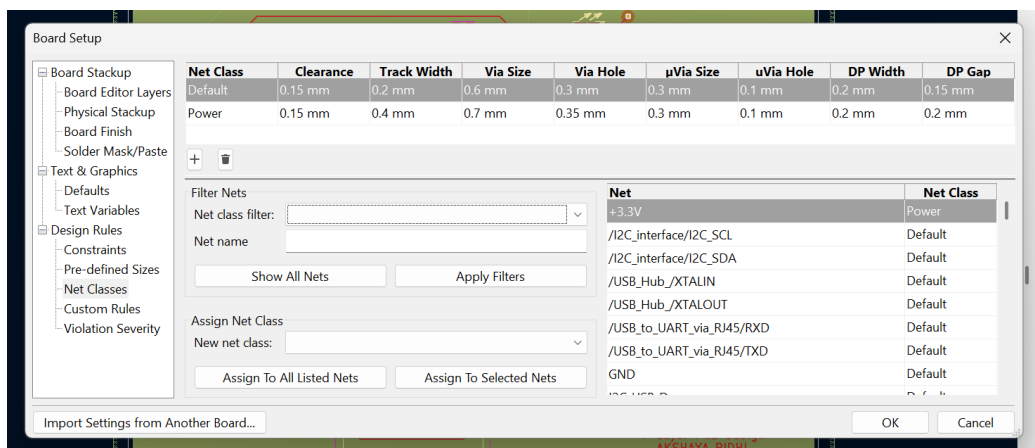


Figure 4.3 Net Classes

Chapter 5: DRC Validation

After completing the component placement, routing, copper zone assignment, and board outline, the final PCB was subjected to Design Rule Check (DRC) to verify the physical integrity of the layout. The DRC was performed using the design constraints defined during Board Setup, including minimum copper clearances, track and via dimensions, pad clearances, board edge clearances, and routing requirements. Priority was given to the USB differential signal paths, power distribution, ground connections, and the interfaces routed to the UART, SPI, and I²C headers.

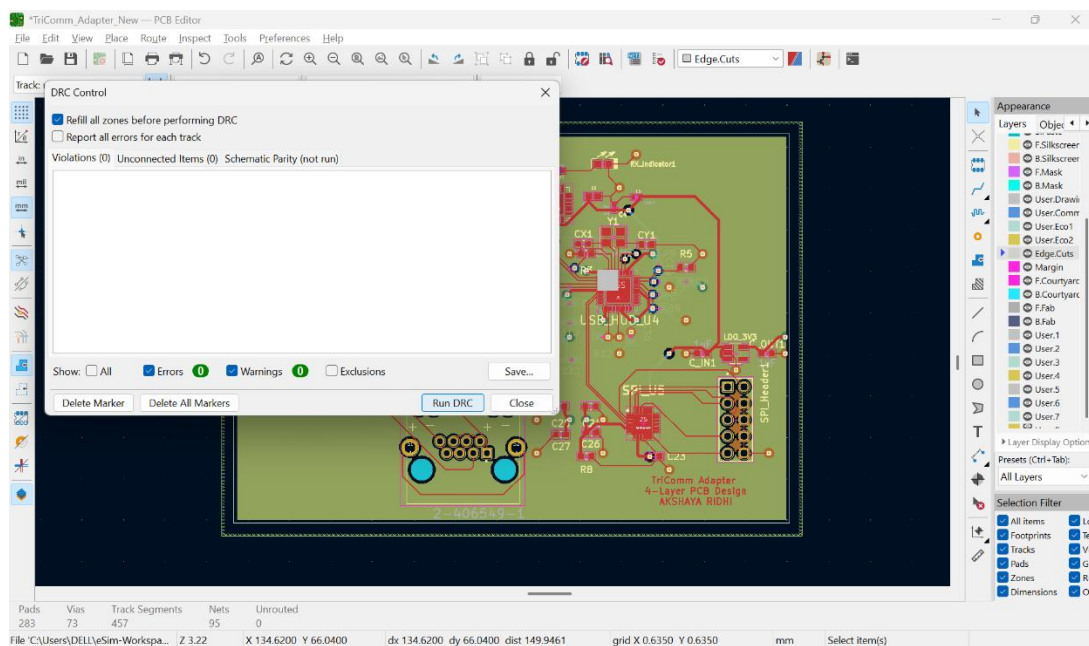


Figure 5.1 Design Rules Check Validation

The final DRC was completed with zero violations, confirming that the PCB layout satisfies all configured physical design constraints. The check verified that there were no unintended copper shorts, insufficient clearances, overlapping or conflicting PCB features, or routing violations. The board outline was also validated as a closed and geometrically valid boundary, ensuring that the PCB can be correctly interpreted for fabrication. The successful result confirms that the placement and routing decisions made during the layout stage are compatible with the defined manufacturing constraints.

The completion of both schematic ERC and PCB DRC validation establishes that the TriComm Adapter has progressed from circuit design to a fully validated PCB implementation. The final board incorporates the USB hub, UART, SPI, and I²C interfaces, regulated 3.3V power distribution, protection circuitry, and multilayer routing within the defined electrical and physical constraints.

Chapter 6: 3D View of PCB

The completed PCB layout was reviewed using the 3D Viewer to verify the physical implementation of the TriComm Adapter before fabrication. The 3D model provides a visual representation of the board with the assigned component packages, connectors, mounting features, and overall board outline, allowing the physical arrangement of the design to be inspected beyond the 2D PCB layout. Along with the successful ERC and DRC validation, the completed 3D view provides a final visual verification of the PCB before fabrication.

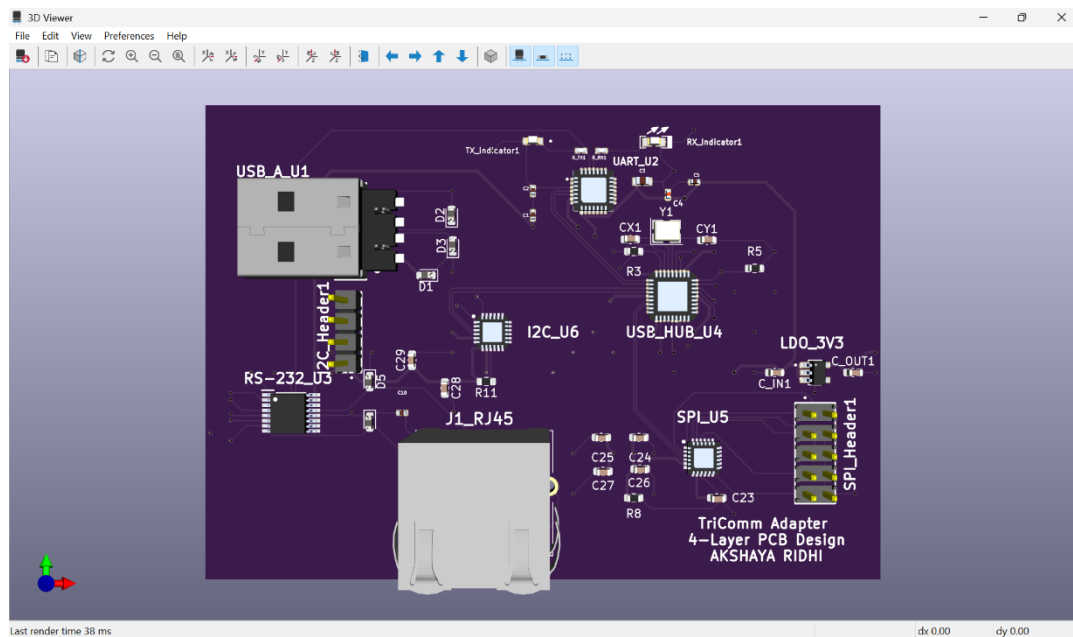


Figure 6.1 3D View Top Layer

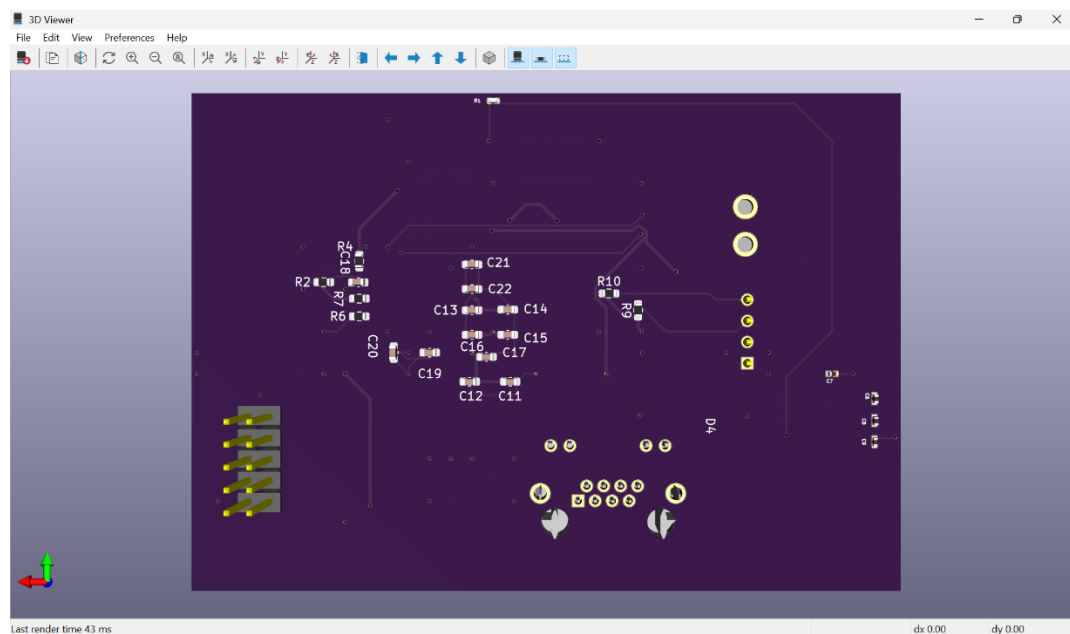


Figure 6.2 3D View Bottom Layer

Conclusion

The TriComm Adapter PCB design successfully integrates USB to UART, SPI, and I²C communication into a single USB powered platform. The project extended the functionality of the reference USB to UART via RJ-45 design by incorporating the USB2513B hub, CP2102 UART bridge, CP2130 SPI bridge, and CP2112 I²C bridge, with an AP2112K regulator providing the required 3.3V supply. The schematic was developed with appropriate power distribution, decoupling, ESD protection, and interface circuitry based on the respective manufacturer recommendations. A four-layer PCB was then designed with dedicated routing and copper planes to support reliable signal and power distribution. Component footprints, routing constraints, net classes, and physical stackup were defined before completing the layout.

The completed design was subjected to ERC and DRC validation, with the final PCB achieving zero violations. The 3D inspection further verified the physical arrangement, footprint assignment, connector placement, and overall board structure. These checks confirm that the schematic and PCB are electrically consistent and that the physical layout satisfies the defined design constraints. The project therefore successfully fulfils the intended PCB design objective for the USB based UART, SPI, and I²C communication. This design task also provided practical experience in schematic development, multilayer PCB layout, high-speed USB routing, power integrity, component footprint selection, design rule management, and hardware verification.

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USB to UART via RJ-45 Module
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USB to SPI
4. [cp2112-datasheet.pdf](#) USB to I2C
5. [AP2112](#) 3V3 LDO Voltage Regulator